

# INTERNATIONAL STANDARD



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**Measurement techniques of piezoelectric, dielectric and electrostatic  
oscillators –  
Part 2: Phase jitter measurement method**



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**Measurement techniques of piezoelectric, dielectric and electrostatic  
oscillators –  
Part 2: Phase jitter measurement method**

INTERNATIONAL  
ELECTROTECHNICAL  
COMMISSION

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## INTERNATIONAL ELECTROTECHNICAL COMMISSION

**MEASUREMENT TECHNIQUES OF PIEZOELECTRIC,  
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International Standard IEC 62884-2 has been prepared by IEC technical committee 49: Piezoelectric, dielectric and electrostatic devices and associated materials for frequency control, selection and detection.

The text of this International Standard is based on the following documents:

|             |                  |
|-------------|------------------|
| CDV         | Report on voting |
| 49/1212/CDV | 49/1243/RVC      |

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 62884 series, published under the general title *Measurement techniques of piezoelectric, dielectric and electrostatic oscillators*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
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## INTRODUCTION

A crystal oscillator as a highly efficient and highly precise source of a frequency oscillation is widely used for fields such as the electronic equipment, communication equipment, measurement equipment and a clock. Also recently, digitalization of these equipments is advancing rapidly. In this situation, the frequency of crystal oscillator requires high precision and high stability and reduction of noise with oscillating phenomenon. A phase jitter is one of the noise characteristic in oscillation characteristic and precise measurement which is needed when shipping a component to a customer.

For advance application in electronic information and communication technology, (e.g. advanced satellite communications, control circuits for electric vehicle (EV)), necessity arises for the measurement method for common guidelines of phase jitter. In these days, measurement method of phase jitter also becomes more important from the electromagnetic influence (EMI) point of view.

This document has been restructured from IEC 60679-1:2007 (third edition) and IEC 60679-6:2011 (first edition). The test methods for oscillators have been separated from IEC 60679-6:2011 into IEC 62884 (all parts). This document covers the phase jitter measurement.



# MEASUREMENT TECHNIQUES OF PIEZOELECTRIC, DIELECTRIC AND ELECTROSTATIC OSCILLATORS –

## Part 2: Phase jitter measurement method

### 1 Scope

This part of IEC 62884 specifies the methods for the measurement and evaluation of the phase jitter measurement of piezoelectric, dielectric and electrostatic oscillators, including dielectric resonator oscillators (DROs) and oscillators using a thin-film bulk acoustic resonator (FBAR) (hereinafter referred to as an "Oscillator") and gives guidance for phase jitter that allows the accurate measurement of RMS jitter.

In the measurement method, phase noise measurement equipment or a phase noise measurement system is used.

NOTE Dielectric resonator oscillators (DROs) and oscillators using FBAR are under consideration.

### 2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60027 (all parts), *Letter symbols to be used in electrical technology*

IEC 60050-561, *International Electrotechnical Vocabulary – Part 561: Piezoelectric, dielectric and electrostatic devices and associated materials for frequency control, selection and detection*

IEC 60679-1:2017, *Piezoelectric, dielectric and electrostatic oscillators of assessed quality – Part 1: Generic specification*

IEC 60469, *Transitions, pulses and related waveforms – Terms, definitions and algorithms*

IEC 60617, *Graphical symbols for diagrams (available at <http://std.iec.ch/iec60617>)*

IEC 62884-1:2017, *Measurement techniques of piezoelectric, dielectric and electrostatic oscillators – Part 1: Basic methods for the measurement*

ISO 80000-1, *Quantities and units – Part 1: General*

### 3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 60027 (all parts), IEC 60050-561, IEC 60469, IEC 60617, IEC 60679-1 and ISO 80000-1 apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>

- ISO Online browsing platform: available at <http://www.iso.org/obp>

## **4 Test and measurement procedures**

### **4.1 General**

The test and measurement procedures are given in Clause 4 of IEC 62884-1:2017 and shall be applied as indicated in 4.2 to Clause 8.

### **4.2 Test methods of phase jitter**

#### **4.2.1 General**

As the measurement method, the phase noise measurement equipment (system) or the specially designed phase jitter measurement equipment shall be used.

Three basic methods are described:

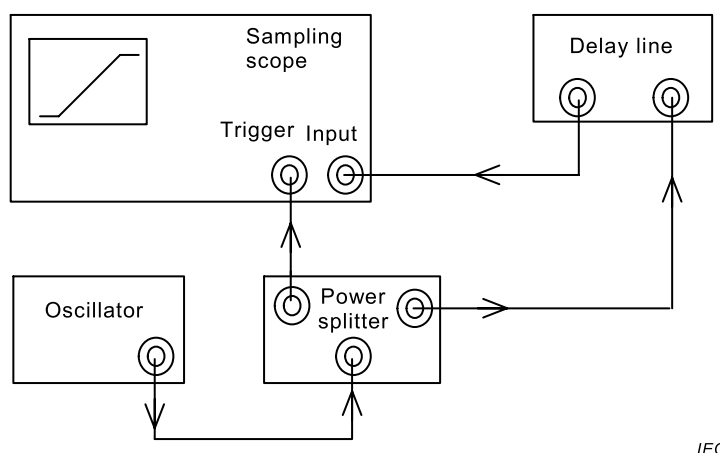
- a) measurement in the time domain by use of a digital real-time or sampling oscilloscope;
- b) measurement in the data domain (BER test set);
- c) measurement in the frequency domain using
  - 1) a phase noise test set, or
  - 2) a jitter and wander test set.

Method c) 1) using a phase noise test set is the recommended measurement method because it allows sufficient accuracy for arbitrary oscillator output frequencies.

- In the measurement of phase jitter and wander of oscillator circuits, attention should be paid to relative measurement reproducibility.
- A user and a manufacturer should deepen understanding through discussion about relative measurement reproducibility.
- Measurement equipment (including software program) should be made clear between a manufacturer and a user through a contract.
- When phase jitter and wander is calculated from phase noise, the range of frequency deviation should be made clear between a user and a manufacturer through a contract.

#### **4.2.2 Measurement in the time domain**

Digital real-time or sampling oscilloscopes with wide bandwidth, fast sampling rates, and large data memories are commercially available (see Figure 1), in some cases with special jitter evaluation software.



**Figure 1 – Phase jitter measurement with sampling oscilloscope**

The time variation of the edges of the clock signal relative to the trigger edge is displayed and stored over a large number (typically thousands) of cycles. Instrument software allows the determination of the peak-to-peak jitter value and a statistical evaluation of its distribution. The sampling oscilloscope method does not allow an accurate evaluation of the spectral content of the jitter. Also, jitter larger than one unit interval (UI) cannot be distinguished.

The measured jitter value is worse than the jitter of the device under test due to the internal jitter of the instrument's clock.

$$J_{\text{DUT}} = \sqrt{(J_{\text{meas}})^2 - (J_{\text{int}})^2}$$

where

$J_{\text{DUT}}$  is the measured jitter;

$J_{\text{meas}}$  is the jitter of the device under test;

$J_{\text{int}}$  is the internal jitter of the instrument's clock.

High stability/low noise Oscillator exhibits a significantly lower jitter than the instrument's clock jitter and trigger stability. Therefore, this technology is currently not suitable for accurate jitter measurement of such Oscillator.

#### 4.2.3 Measurement in the data domain

Bit-error rate (BER) test sets are used for measuring bit-error rate to characterise the overall system performance of a communication subsystem. It is difficult to deduct the contribution of the Oscillator jitter to the system BER. This method also does not yield quantitative jitter performance values for the Oscillator.

#### 4.2.4 Measurement in the frequency domain

##### 4.2.4.1 Methods of phase noise test set

Phase jitter can be tested in the frequency domain using the well-established phase noise test method with a phase locked loop as described in 4.5.25 of IEC 62884-1:2017.

The range of detuned frequency shall be determined by contracts between customers and suppliers after discussion between them. The formula for calculating the RMS jitter from a phase noise is based on the calculation method for the amount of phase jitter shown in Annex A.

For given SDH/SONET applications, the Fourier frequency range ( $f_{\min} \dots f_{\max}$ ) may be selected as described in 3.2.53 of IEC 60679-1:2017. If not specified in the relevant data sheet, the recommended Fourier frequency range is as given by  $f_3$  to  $f_4$  in Table 1.

**Table 1 – Fourier frequency range for phase noise test**

| Oscillator output frequency | $f_0 = f_{\min}$ | $f_3$   | $f_4 = f_{\max}$ |
|-----------------------------|------------------|---------|------------------|
| 1 MHz to < 10 MHz           | 10 Hz            | 10 kHz  | 100 kHz          |
| 10 MHz to < 50 MHz          | 20 Hz            | 20 kHz  | 500 kHz          |
| 50 MHz to < 200 MHz         | 100 Hz           | 50 kHz  | 1,5 MHz          |
| 200 MHz to < 1 000 MHz      | 1,0 kHz          | 200 kHz | 5,0 MHz          |
| 1 000 MHz to < 5 000 MHz    | 5,0 kHz          | 500 kHz | 15 MHz           |
| $\geq 5\,000$ MHz           | 20 kHz           | 2 MHz   | 80 MHz           |

From Table 1, it can be seen that the most stringent requirement applies over the range  $f_3$  to  $f_4$ .

Jitter performance over a frequency-band other than  $f_3$  to  $f_4$  may also be defined.

To compute the phase jitter, the phase noise data  $L(f)$  have to be integrated in the considered frequency ranges and evaluated as follows.

Compute the spectral density of phase fluctuations  $S_\varphi(f)$  from the single-sideband phase noise plot  $10 \log_{10} L(f)$ :

$$S_\varphi(f) = 2L(f)$$

Integrate  $S_\varphi(f)$  over the specified Fourier frequency range  $f_{\min}$  to  $f_{\max}$  to get the mean squared phase jitter in that bandwidth:

$$\langle \Delta\varphi^2(f) \rangle = \int_{f_{\min}}^{f_{\max}} S_\varphi(f) df$$

The mean square phase jitter can be approximated by stepwise integration over the specified Fourier frequency range  $f_{\min}$  to  $f_{\max}$  segmented by  $n$ , for example:

$$\langle \Delta\varphi^2(f) \rangle \approx \sum S_\varphi(f_i) \Delta f_i$$

where

$$\Delta f_i = f_{i+1} - f_i \quad (i = 1 \dots n - 1)$$

with

$$f_1 = f_{\min} \text{ and } f_n = f_{\max}$$

The square root  $\Delta\phi(f)$  of the integral is the effective or RMS phase jitter in radians. It can be converted into degrees, fractions of unit interval (UI), or time (in seconds) by multiplication with the following factor  $k$ :

|       | Degree<br>° | Unit interval<br>UI | Time<br>s      |
|-------|-------------|---------------------|----------------|
| $k =$ | $360/2\pi$  | $1/(2\pi)$          | $1/(2\pi f_c)$ |

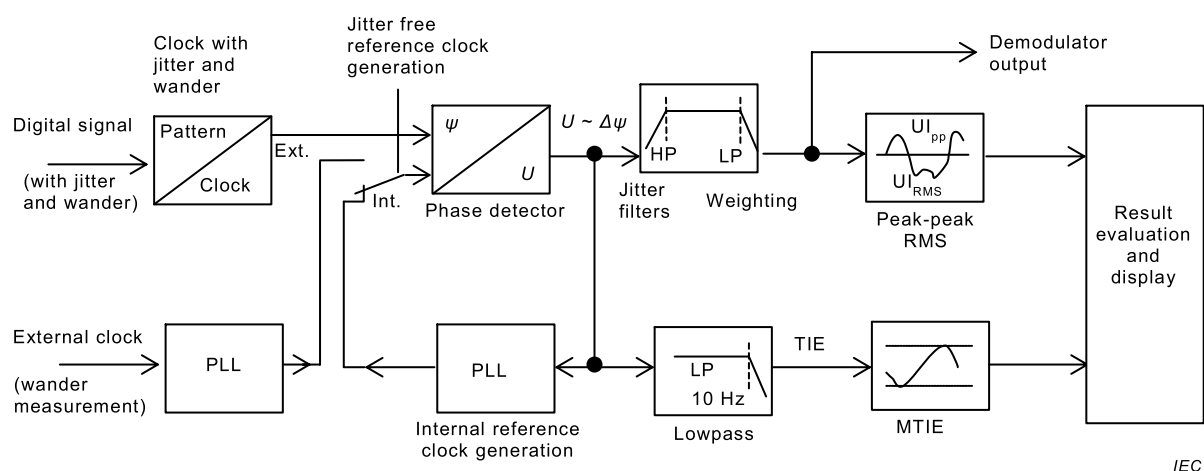
For random jitter, the peak-to-peak value is assumed to be 7 times the value computed above (see 3.2.53 of IEC 60679-1:2017).

Accuracy:

A 1 dB error of the phase noise data  $10 \log_{10} L(f)$  over the full Fourier frequency range causes a jitter inaccuracy of approximately 10 %.

#### 4.2.4.2 Methods of communication analyser

Commercially available communication analyser may be used to measure jitter and wander of clock sources with the method described in ITU-T Recommendation O.172 (see Figure 2). The working principle is similar to the phase noise measurement technique using the quadrature method. Softwares supplied with the test sets deliver directly all characteristic values for jitter and wander in numeric and graphical presentation.



**Figure 2 – Block diagram of a jitter and wander analyser according to ITU-T O.172**

The advantage of these systems over the phase noise test is that a measurement of both RMS and peak-to-peak jitter is possible. The disadvantage is that these systems require an input signal (Oscillator frequency) according to the standard data bit rates for optical communication systems (SONET, SDH) – see Table 2.

**Table 2 – Standard bit rates for various applications**

| <b>SDH</b> | <b>SONET</b> | <b>Bit rate</b><br>Mbit/s | <b>Allowed oscillator frequencies</b> |
|------------|--------------|---------------------------|---------------------------------------|
| -          | OC-1         | 51,84                     | 25,92 MHz, 51,84 MHz                  |
| STM-1      | OC-3         | 155,52                    | 77,76 MHz, 155,52 MHz                 |
| STM-4      | OC-12        | 622,08                    | 311,04 MHz, 622,08 MHz                |
| STM-16     | OC-48        | 2488,32                   | 1 244,16 MHz, 2 488,32 MHz            |
| STM-64     | OC-192       | 9953,28                   | 4 976,64 MHz, 9 953,28 MHz            |

An Oscillator with other output frequencies cannot be tested, which limits the area of application.

NOTE Other applications can have different requirements.

#### **4.2.4.3 Methods of the specially designed measurement equipment**

##### **4.2.4.3.1 General**

The measurement equipment and system shall be the specially designed SONET/SDH measurement equipment using a time interval analyser.

##### **4.2.4.3.2 Measurement items**

The measurement items shall be the RMS jitter and the period (periodic) jitter.

##### **4.2.4.3.3 Number of measurements**

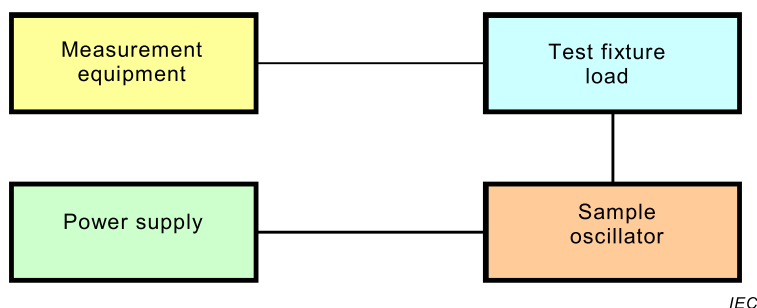
The measurement times shall be determined by contracts between customers and suppliers after discussion between them. The target measurement times shall be 20 000 times or more.

Attention is needed because this device may not meet the requirements of the Oscillator for the following reasons.

- a) The measurable range of the measurement equipment may not meet the frequency of the Oscillator to be measured.
- b) The output voltage of the Oscillator is lower as compared with this device. For this reason, an amplifier is required, and the necessity of evaluating the phase jitter of the amplifier arises.
- c) The realization of square waves, such as CMOS, LVDS, and LVPECL, is difficult because harmonics components decrease in the frequency bands exceeding 300 MHz. For this reason, the signal waveforms become sine waves, clipped-sine waves and the like. It is difficult to analyse them by the specially designed SONET/SDH measurement equipment, and thus a decrease in measurement accuracy is possible.

##### **4.2.4.3.4 Block diagram of the measurement**

A representative block diagram is shown in Figure 3. A practical block diagram is utilized as modified forms of Figure 1.



**Figure 3 – Equivalent block diagram**

### 4.3 Input and output impedances of the measurement system

The load impedance of the Oscillator widely ranges from 5  $\Omega$  to 100 M $\Omega$ . The parts to be applied are the types shown below. However, since numerous demands are made by customers, the values of this load impedance are infinite.

- a) capacitor only;
- b) resistor only;
- c) both capacitor and resistor;
- d) compliment output with bias.

Here, since the measurement system is unified into 50  $\Omega$ , the input-output impedances of measurement systems shall be 50  $\Omega$ . For this reason, the load impedance of the Oscillator shall also be 50  $\Omega$ .

The oscillation output voltage changes depend on the load impedance of the Oscillator. For this reason, the thermal noise of load circuits also changes.

As a result, since the amount of phase jitter changes, a recommendation is presented to suppliers and customers, when adopting any load impedance other than 50  $\Omega$ , to conduct a detailed study and examination and to determine the impedance by contract.

### 4.4 Measurement equipment

#### 4.4.1 General

The specification required for the measurement equipment is described in 4.4.2 to 4.6 without any necessity of adhering to this document. The adoption of measurement equipment which satisfies sufficiently the requirements of the Oscillator is important.

#### 4.4.2 Jitter floor

The jitter floor shall take values smaller by one digit as compared with the phase jitter demanded for the Oscillator.

#### 4.4.3 Output wave form

The output waveforms shall be CMOS, LVDS, LVPECL, clipped-sine waves, sine waves, etc.

NOTE CMOS, LVDS, and LVPECL originally refer to the type of devices and not a waveform *per se*. However, they are also used as the terms showing the waveforms and are, therefore, described as the type of output waveforms in this document.

#### **4.4.4 Output voltage**

The output voltage shall be 350 mV or more.

#### **4.5 Test fixture**

The specification demanded for measurement implements is shown below.

- a) Connection between the Oscillator to be measured and measurement implements  
The application of sockets, connectors, screws, clips, and the like may be allowed. In addition, the Oscillator to be measured and the measurement implements shall be ensured to be mechanically and electrically connectable.
- b) Compatibilization of oscillators to be measured and measurement implements  
The Oscillator to be measured and the measurement implements shall be capable of being earthed.
- c) Although the load impedance may not be built in, a recommendation is presented to use measurement implements having the load impedance built therein in order to reduce influences on the phase jitter of the Oscillator to be measured from a thermal noise or the like from the load impedance.

#### **4.6 Cable, tools and instruments, and so on**

As for a cable, the double-shield type of a 50  $\Omega$  system shall be used. The cable shall be as short as possible. As connectors, the 50  $\Omega$  system shall be used. It is recommended that SMA or N-type connectors be used.

NOTE This measurement system is a 50  $\Omega$  system. This is the definition of the viewpoint of a measuring method. The actual load impedance of the Oscillator cannot say it as a 50  $\Omega$  system. When a measurement system is not a 50  $\Omega$  system, user and supplier permit mutually use of the measurement system which is not a 50  $\Omega$  system. User and supplier can define a new measurement system by a contract.

### **5 Measurement and the measurement environment**

#### **5.1 Set-up before taking measurements**

Attention should be paid to the following matters.

- a) The entire measurement system and the Oscillator to be measured shall be installed in a measurement chamber at least 2 h previously.
- b) The measurement equipment shall be set to operate for 2 h or more.
- c) The frequency stability of clock signals in the measurement equipment shall be verified to be smaller than, or equivalent to, the frequency stability of the Oscillator to be measured.
- d) The power supply voltage of the Oscillator to be measured and the measurement equipment shall be verified to be set to the AC voltage and the DC voltage as requested.
- e) Restrictions shall be provided for the operation of surrounding electronic devices so as not to produce an electronic noise from the surrounding.

#### **5.2 Points to be considered and noted at the time of measurement**

No vibration of the measurement system shall be caused. No movement shall be caused. No shifting of the cable position shall be made.

#### **5.3 Treatment after the measurement**

It is preferable not to disassemble the measurement system after performing measurements. Periodical inspection and calibration of the measurement equipment should be ensured.



## **6 Measurement**

### **6.1 Reference temperature**

The reference temperature shall be  $+25\text{ °C} \pm 5\text{ °C}$ .

### **6.2 Measurement of temperature characteristics**

Only the Oscillator to be measured shall be immobilized in the precisely variable temperature bath as appropriately selected, and the temperature characteristics shall be measured. No vibration shall be caused.

### **6.3 Measurement under vibration**

Only the Oscillator to be measured shall be fixed to the shaker as appropriately selected and caused to vibrate. No vibration of the measurement equipment shall be caused.

### **6.4 Measurement at the time of impact**

Only the Oscillator to be measured shall be fixed to the impact machine as appropriately selected to apply impact thereto. Moreover, no shock wave or no vibration accompanied with the impact shall be provided for the measurement equipment.

In addition, this testing is not realistic because the impact period of time is shorter than the measurement period of time. If this testing is performed, a recommendation is given to suppliers and customers to conduct a detailed study and examination and to determine the measurement by contract.

### **6.5 Measurement in accelerated ageing**

Only the Oscillator to be measured shall be set to the temperature and time based on the specification in the temperature bath as appropriately selected, and then caused to immobilize, and thus the accelerated ageing shall be measured.

## **7 Other points to be noted**

Consideration shall be taken so as to obtain the measurement results understandable to suppliers and customers by eliminating any possibility that an electronic noise can be involved in the measurement system from the supply source line through paying attention also to the phase jitter of the devices applied to the measurement system, or to be applied around the system.

## **8 Miscellaneous**

With regard to the amount of phase jitter of the Oscillator, as well as modules that have a multiplication function or a division function based on these oscillators, customers and suppliers shall conduct a detailed study and examination, and determine this by contract.

## Annex A (normative)

### Calculation method for the amount of phase jitter

#### A.1 General

Annex A gives the method of calculating the amount of phase jitter from phase noise measurement results.

#### A.2 Explanation

When the amount of phase jitter is calculated from the phase noise measurement results, the RMS jitter can be obtained. The details are described below.

If a spectrum analyser or a phase noise measurement system is used, the phase jitter can be analysed as to the frequency components which can be used for the cause analysis of the phase jitter. According to the measurement of the phase jitter by the phase noise measurement system, the ultra-low amount of phase jitter, which cannot be measured by other jitter measurement methods, can be measured, and thus the phase noise measurement system is suitable for evaluating highly stable devices such as the Oscillator. With regard to the signals of the Oscillator, various types of signal waveforms such as sine waves and square waves are requested by customers. Among them, as for the sine wave signals, the application of the phase noise measurement system is theoretical and appropriate. However, as for the square wave signals, although error-increasing factors are involved, since any other method capable of firmly measuring the ultra-low amount of phase jitter has not yet been found, the phase noise measurement system shall be applied.

In general, when the measurement results of an SSB phase noise of the Oscillator are viewed, the offset frequency in the horizontal axis is described such as 10 Hz to 1 MHz, 1 Hz to 1 MHz, and 1 Hz to 10 MHz in many cases. In particular, for the offset frequency of 10 kHz or more as the floor level, the offset frequency is described as 1 MHz or 10 MHz. Such offset frequency is obtained because filters are provided in the measurement equipment.

On the other hand, as for the phase jitter, since such filters are not required, the measurement values can be obtained regardless of the offset frequency. Therefore, no complete coincidence can be maintained to be provided for the phase noise measurement values and the phase jitter measurement values. However, in the case of the Oscillator having the ultra-low amount of phase jitter, the phase noise measurement values and the phase jitter need to be correlated, and, therefore, the phase noise and the phase jitter are used for convenience.

#### A.3 Relations between phase noise and phase jitter

When phase modulations are demodulated by a phase detector (converting phase fluctuations into voltage fluctuations), the relationship between phase and voltage can be expressed by Formula (A.1), wherein  $K_\phi$  is a constant, and the unit is  $K_\phi$  (V/rad).

$$\Delta V_{\text{out}} = K_\phi \times \Delta\phi \quad (\text{A.1})$$

When the converted phase fluctuations are measured by a spectrum analyser, the relationship can be expressed by Formula (A.2):

$$\Delta V_{\text{rms}}(f) = K_\phi \times \Delta\phi_{\text{rms}}(f) [V] \quad (\text{A.2})$$

wherein, if  $S_{\text{v}_{\text{rms}}}(f)$  is defined as the spectral density function of the voltage fluctuations (output fluctuations of the phase detector) as measured, the spectral density function of the phase fluctuations can be expressed by Formula (A.3):

$$\begin{aligned}
 S_{\phi}(f) &= \frac{(\Delta\phi_{\text{rms}}(f))^2}{B} \\
 &= \frac{(\Delta V_{\text{rms}}(f))^2}{K\phi^2 \times B} \\
 &= \frac{S_{\text{v}_{\text{rms}}}(f)}{K\phi^2} [\text{rad}^2 / \text{Hz}]
 \end{aligned} \tag{A.3}$$

where

$B$  is the bandwidth of the spectrum analyser.

When the results are converted into the SSB phase noise as shown below, the SSB phase noise can be expressed by Formula (A.4):

$$L(f) = \frac{S_{\phi}(f)}{2} \tag{A.4}$$

where

$S_{\phi}(f)$  is a dB value relative to 1 radian, and also the power spectral density function of the phase fluctuations;

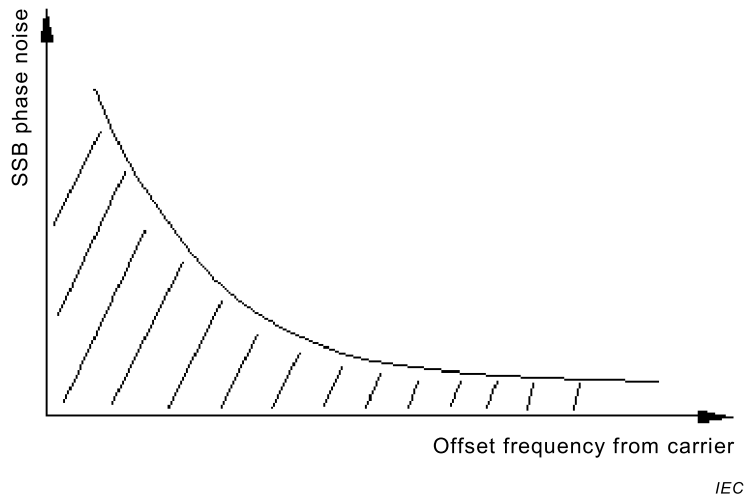
$L(f)$  is the single sideband (SSB) phase noise.

A total phase deviation in the designated band, namely, the phase jitter, can be expressed by Formulae (A.5) and (A.6):

$$\Phi = \sqrt{\int_A^B S_{\phi}(f) \cdot df} [\text{rad}] \tag{A.5}$$

$$\Phi = \sqrt{\int_A^B 2 \cdot L(f) \cdot df} [\text{rad}] \tag{A.6}$$

Therefore, the shaded parts (area of SSB phase noise) shown in Figure A.1 can be referred to as the phase jitter. This area accords with the square of the RMS jitter. Here, if the offset frequency range is different, the phase jitter calculation value becomes different. Since the fact is a shortcoming of this method, attention should be paid when calculating the phase jitter from the SSB phase noise.



**Figure A.1 – Concept diagram of SSB phase noise**

#### **A.4 Commentary on theoretical positioning of phase jitter**

Frequency stability was compiled into a single work by IEEE in 1966 [3]. Then, the definition is applied to atomic oscillators, crystal oscillators, as well as electronic systems for telecommunication, information, audio-visual, and the like.

The conventional crystal oscillators and electronic systems have analogue systems with exception of a part, and the signal waveforms are sine waves. Therefore, the short-term frequency stability as one field of the frequency stability is measured as the phase noise or Allan variance. Recently, digitization of electronic systems is progressing. Under such circumstances, the short-term frequency stability has been measured as the phase jitter.

On the other hand, Oscillators are analogue-type electronic devices. For the Oscillator, the signals having square waves or waveforms similar thereto are demanded by users for facilitating to be assembled into the electronic systems. Naturally, for the short-term frequency stability, the measurement as the phase jitter is frequently demanded by users.

#### **A.5 Description**

##### **A.5.1 General**

The phase jitter of an Oscillator means an electronic noise of signal waveforms in terms of time. On the other hand, the phase jitter is described as a jitter in which the frequency of signal deflection exceeds 10 Hz and as a wander in which the frequency is 10 Hz or less.

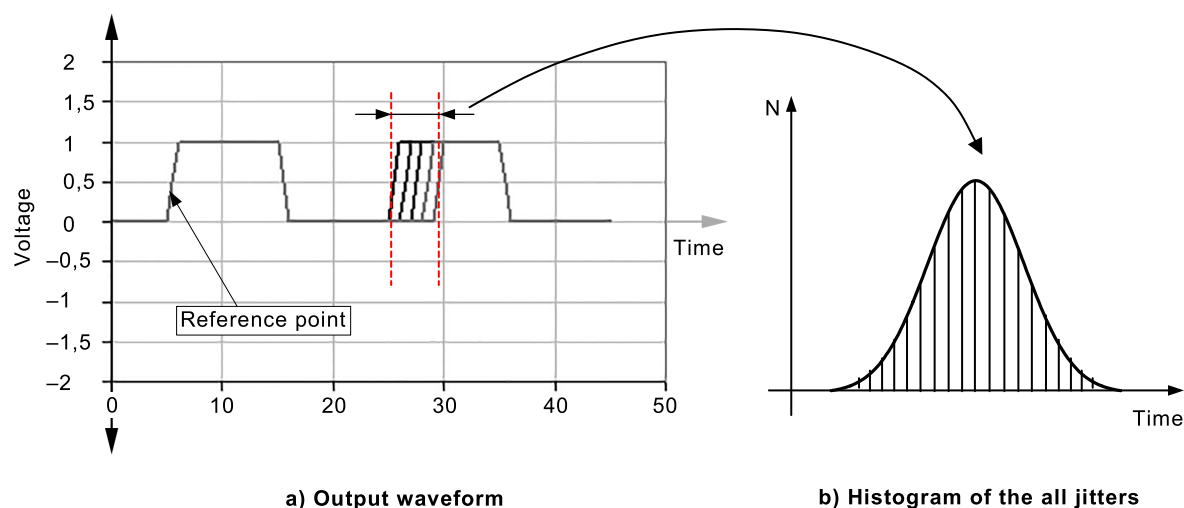
It is difficult to observe the wander of an Oscillator. The wander is a phenomenon which is confirmed in electronic parts such as optical cables susceptible to expansion and contraction even by a small amount of temperature changes. Therefore, the wander is generally not discussed in the Oscillator. In this document also, the phase jitter is targeted only to the jitter.

As for signals, an ideal one cycle ( $t$ ) is inversely proportional to a frequency ( $f$ ). More specifically, the relation is expressed by Formula (A.7):

$$t = \frac{1}{f} \quad (\text{A.7})$$

Actually, the cycle is varied by receiving various influences. This phenomenon is the phase jitter and can be confirmed by thickening of edges of waveforms when using oscilloscopes or the like. As the method of measuring and evaluating such phase jitter, statistical measurement techniques are utilized. This phenomenon is shown in Figure A.2 a). The numerical values in Figure A.2 a) are treated as a symbol. The position of 0,5 of signal waveforms is defined as a reference point in the vertical axis, and the edges of the reference point are defined to be not varied. When attention is paid to the edges after one cycle, every time when the signals repeatedly move on the screen of CRT in the lateral direction, the edges after one cycle are not reproduced. Then, plurality edges have become to exist. This phenomenon is induced when repeatedly measuring the signals, and referred to as the phase jitter.

This phase jitter is treated as a normal distribution – see Figure A.2 b). Then, when analysed, the phase jitter can be divided into several types of properties. More specifically, the phase jitter is classified in several types. In this document, the phase jitter is classified in the seven types as described below. In A.5.2 to A.5.8, these properties and the cause systems are made clear.



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**Figure A.2 – Voltage versus time**

### A.5.2 RMS jitter

The RMS jitter is the phase jitter which comes to have the normal distribution shown in Figure A.3. The RMS jitter is a standard deviation obtained on the basis of statistical treatments and defined as a  $1\sigma$  portion.

According to statistics, any measurement data is meant to exist in  $1\sigma$  at a probability of 68,26 %. Therefore, when the measurement times are 10 000, approximately 6 826 pieces of the measurement data are considered to be contained. On the contrary, 31,74 % (3 174 pieces) of the measurement data are indicated to be outside the plus and minus sides of  $1\sigma$ . If the data outside the definition is considered to be errors, 31,74 % can be considered to be the error rate.

### A.5.3 Peak-to-peak jitter

The peak-to-peak jitter is the phase jitter which comes to have the normal distribution shown in Figure A.3. The amount of phase jitter of one cycle is totalized and statistically treated on the base point of the reference point of phase jitter shown in Figure A.2. In this case, the amount of phase jitter is assumed to provide the normal distribution.

The difference between the maximum value and the minimum value (namely, change width) is referred to as the peak-to-peak jitter. Since the jitter values become larger as the measurement times are increased, the jitter also becomes the total jitter as described later. This term comes on when negotiating specifications between customers and Oscillator makers.

Since the peak-to-peak jitter or the RMS jitter indicates the amount of phase jitter in the measurement times thereof, the jitter indicates operating conditions of measurement samples in a short period of time. Moreover, the jitter has values effective only to an ideal normal distribution (Gaussian distributions), and the effectiveness can be maintained to be low when considered for the cases of non-Gaussian distributions having distorted distributions such as binomial distributions and chi-square distributions. Accordingly, when applying the peak-to-peak jitter or the RMS jitter, the measurement times are required to be clearly defined by contracts between customers and supplier sides.

#### **A.5.4 Random jitter**

The random jitter is shown in Figure A.4. The random jitter represents unpredictable phase jitter components.

The random jitter naturally and inductively occurs as influenced by the characteristics, thermal noise, and the like originally involved in the measurement equipment *per se* or Oscillator. Furthermore, the random jitter has characteristics that the distribution width of measurement values becomes larger (namely, boundless characteristics) as the observation period of time becomes longer. Therefore, the distribution chart can be considered as an ideal normal distribution. Moreover, the random jitter is determined as a standard deviation based on the distribution chart obtained by the measurement of phase jitter. Accordingly, in the case of an Oscillator, the random jitter may become the amount of jitter equivalent to the RMS jitter. Moreover, since the random jitter becomes the amount of jitter of the measurement equipment per second, the random jitter is one of the measures for judging applicability to measuring the phase jitter of an Oscillator.

#### **A.5.5 Deterministic jitter**

The deterministic jitter occurs by various factors of regularity (circuit designs, electromagnetic induction, or induced from external environment), and has characteristics inasmuch as the change width of distribution has a boundary and thus can be expressed by the parts sandwiched between right and left random jitters. On the other hand, the components forming the deterministic jitter include the period jitter or periodic jitter and the data-dependent jitter.

#### **A.5.6 Period (periodic) jitter**

The period jitter or periodic jitter shows variations of timings of multiple cycles consecutively provided such as two cycles and three cycles. The period jitter or periodic jitter can be determined by grasping the relationship with the RMS jitter between the multiple cycles and each cycle and thus grasping whether or not periodic irregularities appear. As for the periodic components of this jitter, such components are considered as an electronic noise caused by the power supply and cross-talk from electronic parts around the Oscillator to be measured, and further from cores in the vicinity in the case of IC.

If the fast Fourier transform (FFT) can be executed, the frequency as the cause clearly appears as a spectrum. Although this jitter is naturally required to be considered for the Oscillator, it is difficult to detect the jitter by using measurement equipment in general.

#### **A.5.7 Data-dependent jitter**

The data-dependent jitter is considered to be jitter components due to duty cycle distortion and inter symbol interference, and is negligible for the Oscillator.

### A.5.8 Total jitter

The total jitter is defined as the jitter obtained by totalizing all of the jitters.

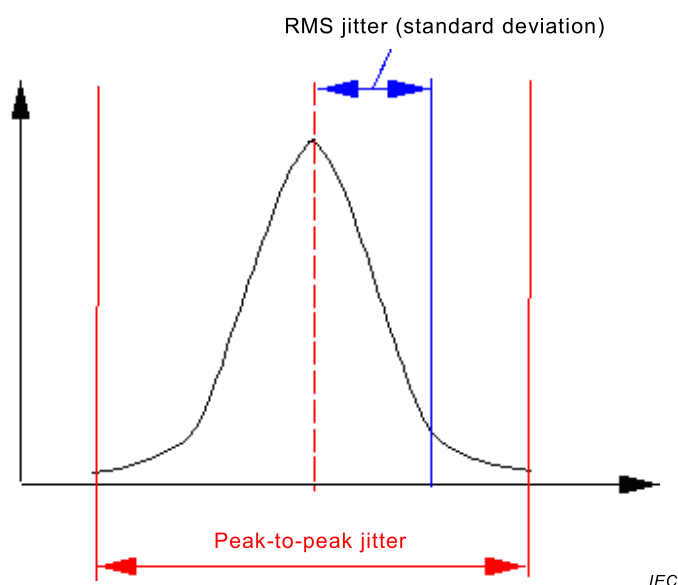
## A.6 Points to be considered for measurement

### A.6.1 Measurement equipment

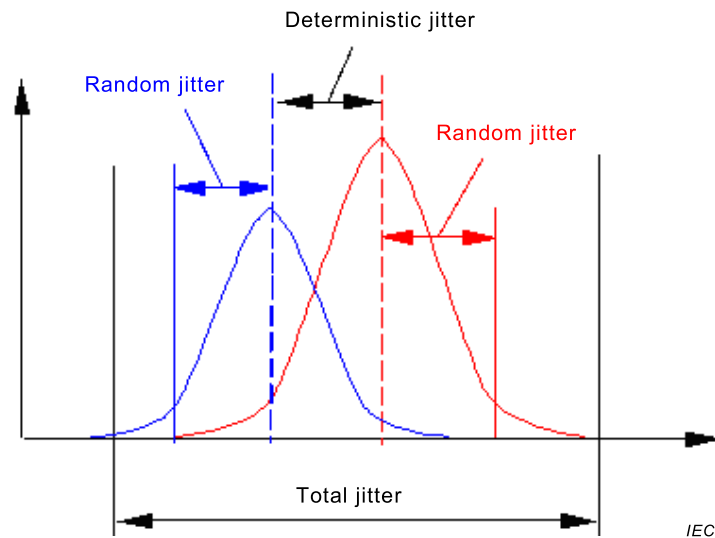
For the Oscillator, requests of infinite variety are provided by customers. The output waveforms are not limited to square waves. The demands for output voltage as small as not applicable to the measurement equipment can also be provided.

Since the Oscillator have an ultra-low noise, such a case can be experienced that the amount of jitter of the measurement equipment per se is detected. Therefore, for the amount of jitter of the measurement equipment per se, the measurement equipment shall have the jitter floor smaller by one digit as compared with the amount of jitter of assumed Oscillator. Moreover, the frequency range and the output waveforms are requested to be applicable not only to square waves but also to sine waves.

Since measurement equipments in general are provided with the specification of a degree applicable to digital electronic systems, a sufficient study is required for adopting the measurement equipment for oscillator purpose.



**Figure A.3 – Explanatory diagram of the amount of jitter applied to RMS jitter**



**Figure A.4 – Explanatory diagrams of random jitter, deterministic jitter, and total jitter**

The following points shall be considered for each measurement system.

- a) In the case of digital oscilloscopes, no appropriate measurement equipment for such oscillator purpose is found.
- b) When applying digital oscilloscopes to the phase jitter measurement, a recommendation is presented to select the measurement equipment and the measurement system capable of being sufficiently verified and confirmed, and to be determined by contract between suppliers and customers.
- c) In the case of specially designed measurement equipment to which a time interval analyser is applied, in a part of measurement equipment, the type having such shortcomings exists that the random jitter of jitter floor is equivalent to, or larger than, compared with an Oscillator, that the application of sine waves is difficult, that the low frequency cannot be applied to the range of such Oscillator, and that the output voltage is low and an amplifier is required. Therefore, sufficient consideration is required for selecting the type of specially designed measurement equipments.
- d) The phase jitter can be calculated from phase noise measurement values by using the phase noise measurement equipment or the measurement system. In this case, the detuned frequency shall be determined by contract between suppliers and customers. When the detuned frequency becomes outside the range of the phase noise measurement equipment or the measurement system, in particular, when the upper limit of the detuned frequency becomes a floor level, consideration shall be taken not to produce a misunderstanding between customers and suppliers by defining that the voltage of floor level is flat according to contract between them.

Within the range investigated during the first stage, no devices satisfying the requirements are found among sampling oscilloscopes and specially designed measurement equipments. However, since information is obtained that a part of specially designed measurement equipment satisfying the requirements has been put on the market, the specially designed measurement equipment is taken in the standard.

#### **A.6.2 Factors of measurement errors**

With regard to the Oscillator, the factors of phase jitter measurement errors are shown as follows.

- a) Power supply

The power supply is required for driving the Oscillator. If unstable power supply is used, the unstable power supply is observed as converted into the jitter, and therefore the use of



power supply having a sufficiently low noise is desirable. Since losses are produced for the wiring cable between power supply terminals and Oscillator or amplifiers, and contact resistance is produced, the amount of phase jitter can be increased from this part.

b) Test fixer and load

The load is formed of resistors and fixed capacitors. Since the resistors exist, generation of electronic noise cannot be avoided. The possibility of playing a role of collecting the electronic noise as an antenna can be exemplified.

c) Amplifier (when using an amplifier at the time of measurement)

The amplifier is formed of electronic parts including active elements and resistors. Therefore, generation of electronic noise cannot be avoided.

d) Cable

The cable includes losses therein and therefore is an electronic noise generation source. Since the reflectance changes as the impedance changes through the change of length caused by temperature characteristics, the change may be misread as the wander. An electronic noise due to contact change of connectors can be provided. The possibility of playing a role of collecting the electronic noise as an antenna can be exemplified.

e) Input-output impedances of measurement system

The load impedance of the Oscillator widely ranges from  $5\ \Omega$  to  $100\ \text{M}\Omega$ . The parts used for the load impedance include the following three types.

- 1) only capacitor;
- 2) only resistance element;
- 3) combined use of capacitor and resistance element.

In the case of 1) only capacitor phase jitter measurement values can be neglected. In the case of 2) only resistance element, and in 3) combined use of capacitor and resistance element, attention is needed because the phase jitter measurement values cannot be neglected by the thermal noise from the resistance element.

f) Measurement of phase jitter for frequencies exceeding 1 GHz

In general, the waveforms of signals (including demodulated signals) exceeding 1 GHz are modified sine waves. Therefore, attention is needed because the amount of phase jitter, which suppliers and customers intended, may be difficult to obtain by sampling oscilloscopes or specially designed measurement equipments.

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